

Darshan C

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PROFILE

Embedded Systems Engineer with strong foundation in EEE, specialising in development of IoT solutions, robust firmware for Espressif, Arduino Devices and Designing custom PCB's using KiCad, Proven track record in creating seamless Human Machine Interfaces using Flutter to bridge gap between hardware and user.

EDUCATION

B.E Electrical and Electronics Engineering | Vidya Vikas Institute of Engineering and Technology | 2022 – 2026 | 7.27 CGPA

SKILLS

Technical: C, C++, Embedded C, Python, Flutter, EDA, Multi-Layer PCB Design, CAED, Arduino Framework, ESP-IDF, FreeRTOS, ESP32, ESP8266, ATmega328P, Basic Assembly, Power Electronics, UART, I2C, SPI, UDP, TCP, UI/UX Design

Tools: Platform IO, Android Studio, KiCad, Arduino IDE, MATLAB, Solid Edge, Blender, NI Multisim, Keil u Vision, VS, MS Office

CERTIFICATIONS

- UPCITI – Workshop on Power Electronics Drives and Electric Vehicles – April'25
- IJRSET – Certificate of Publication on App-Controlled Smart Door Lock System – December'25

EXPERIENCE

PAT Technologies Private Limited (Edutainer)

Bangalore, India

UX Design, Game Design Intern

Feb 26 – May 26

- Designed intuitive Human-Machine Interfaces (HMI) for interactive systems to improve Hardware Accessibility.
- Collaborated on the architectural design of interactive game systems using unity.
- Conducted rigorous usability testing and iterative design cycles to ensure seamless software and user interactions.

PROJECTS

InvisiLock – An Emergency Aware Smart Door Lock System

- Conventional Smart Lock Can Malfunction, Irresponsive to Emergency due to Exposed Exterior Electronics.
- ESP32, Platform IO, ESP-IDF, UDP, KiCad, Flutter, Experience in Multi-Layer PCB Design, Prototyping & Debugging.
- Improved durability and security, Reliable Multi-Mode access, User Safety and rescue significantly increased.

Smart Solar Fencing System – An Autonomous Shock Intensity Varying System

- Conventional Electric Fence gives high intensity shocks, Designed Auto/Manual iterative shock intensity.
- ESP32, Platform IO, Arduino framework, Flutter, UDP, Experience in Prototyping & Debugging.
- Achieved safer, more controlled shock intensity reducing risk to human & animals while having effective protection.

MediWatch – A Health Monitoring and IV Control System

- Lack of continuous, accurate patient monitoring, Developed Mediwatch to Remotely monitor Patient Conditions.
- ESP32, SPI, Platform IO, Arduino Framework, Flutter, UDP, Kicad, Multi-Layer PCB Design, Prototyping & Debugging.
- Achieved accurate real-time health monitoring and IV Flowing.

ACHIEVEMENTS

- 1st Place – Electrovananza'26: Awarded at the National Level Project Competition hosted by SJCE.
- 1st Place – V-VISION 2025: Secured top honours at National Level Project Expo hosted by VVIET.
- 3rd Place – VRIDDHEE 2025: Recognised at the State Level Project Competition hosted by VVCE.
- 3rd Place – NEWENTURE 2025: Awarded for high-impact pitch in the idea pitch contest hosted by VVIET.